INCH-POUND

MIL-T-52035C(ME)
25 April 1991
SUPERSEDING
MIL-T-52035B(ME)
19 December 1979

MILITARY SPECIFICATION

THEODOLITES, SURVEYING, DIRECTION, FIRST-ORDER, WITH ACCESSORIES, TRIPOD, AND CASES

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the night-illuminated, direction-type, first-order theodolites with accessories, tripod, and cases.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC FSC 6675

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

A-A-208	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).
W-L-111	- Lamp, Electric Incandescent Miniature, Tungsten-Filament.
TT-F-325	- Filler, Engraving, Stamped Marking.
TT-V-121	- Varnish, Spar, Water Resisting.
CCC-C-419	- Cloth, Duck, Cotton, Unbleached, Plied-Yarns Army and Numbered.
NNN-P-40	- Paper, Lens.

MILITARY

MIL-B-18	- Battery, Dry.
MIL-B-18/9	- Battery, Dry, BA-30.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-H-639	- Hooks (for Belts, Straps, and Equipage Items).
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-S-3534	- Surveying Instruments and Accessories, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-R-14328	- Rubber Sheet: Synthetic Medium Soft, General Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-F-46032	- Fungus Resistant Treatment for Sandbags; Copper Processes.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-731	- Quality of Wood Members for Containers and Pallets.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-810	- Environmental Test Method and Engineering Guidelines.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government publication. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the specification.

NATIONAL BUREAU OF STANDARDS (NBS)

Special Publication 374 - Method for Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the National Technical Information Service, 5285 Port Royal Road, Springfield, VA 22161.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

Rules for the Use of Volatile Organic Compounds (VOC).

(Application for copies should be addressed to the South Coast Air Quality Management District, Attn: Public Affairs Department, 9150 Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specifications sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The theodolite shall be a direction-type instrument capable of first-order accuracy (see 6.5). The theodolite shall consist essentially of an alidade, horizontal and vertical circles, a microscopic optical system for reading the circles and scales, and optical plummet, a base, and a leveling head. The alidade shall contain the telescope, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

vertical circle, and the circle-reading micrometer and eyepiece. The base shall contain the horizontal circle, the vertical axis system, and the leveling head.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The theodolite shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the theodolite may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the theodolite may be newly fabricated from recovered materials to the maximum extent practicable, provided the theodolite produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the theodolite.

3.4 Environmental conditions.

3.4.1 Operating temperature. The theodolite, levels, tripod, cases, and accessories shall operate as specified in 4.5.2.6 and 4.5.2.8, without damage, when subjected to ambient temperatures from 125 ^{0}F (51.7 ^{0}C) to -40 ^{0}F (-40 ^{0}C).

3.4.2 Storage temperature. The theodolite, tripod, levels, cases, and accessories shall not be damaged by storage under the following conditions:

Air temperature:

High - 155 ^{0}F (68.3 ^{0}C). Time period, 4 hours.
Low - -50 ^{0}F (-45.6 ^{0}C). Time period, 6 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

d. Transit the telescope to the reverse position and sight

3.4.3 Solar radiation. The theodolite shall operate as specified in 4.5.2.10, and shall not be damaged or show evidence of peeled, cracked, or blistered paint by full impact of solar radiation of 360 British thermal units per square foot per hour (Btu/ft^2/hr) at an air temperature of 125 °F (51.7 °C).

3.4.4 Rainfall. The theodolite, metal carrying case, and transporting case shall show no signs of water penetration when subjected to rainfall at a rate of 3 inches, ±1 inch (76.2 millimeters (mm), ±25.4 mm), per hour, as specified in 4.5.2.11.

3.4.5 Sand and dust. When the theodolite and metal carrying cases are subjected to blowing sand and dust, as specified in 4.5.2.12, the theodolite shall not be damaged; shall not show sand and dust penetration in the telescope, in the optical system, and on bearings; and shall not show lens abrasion. The metal carrying case shall not show sand or dust penetration.

3.4.6 Vibration. The theodolite and the metal carrying case shall not be damaged by vibration when tested as specified in 4.5.2.13.

3.4.7 Shock. The theodolite, tripod, levels, and cases shall not be damaged mechanically or electrically by shock when tested as specified in 4.5.2.14.

3.4.8 Humidity. The theodolite shall not be damaged and shall operate within the accuracy specified in 4.5.2.15, when subjected to relative humidity of at least 95 percent at air temperature varying between -25 °F (-31.7 °C) and 110 °F (43.3 °C).

3.5 Telescope. The telescope shall be internal focusing and dust and moisture resistant (but not pressure tight). The telescope shall be centered over the vertical axis and, with the precision of the focusing movement of the negative lens, shall not change more than 40 seconds of arc over a focusing range of 62 feet (18.9 meters (m)) to infinity.

3.5.1 Optical system. The optical system shall produce good definition throughout the entire focusing range and shall be free of internal reflection, spherical and chromatic aberration, coma astigmatism, and distortion. The field of view shall be substantially flat with no marked curvature within one-half the diameter of the field of view. The principal axis of the lens system shall be in alinement and coincident with the collimation of the telescope of adjustment.

3.5.1.1 Coated optics. Except for the registered face of the reticle, the internal lens-air surfaces of the telescope shall be coated with a reflection reducing film. The reflectivity and absorption of the film shall not exceed 0.5 percent in accordance with MIL-C-675.

3.5.1.2 Mountings. Mountings for all prisms and lenses used in the optical system by which the circles are read shall not shift. Optical components

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

not as specified (see 3.7.1.1) 126 Mildew resistance

shall be free from strains within the mountings. Metal surfacing on the optical components shall be free from distortion and shall be made of material which shall not tarnish.

3.5.2 Telescope eyepiece. The telescope shall be fitted with two standard eyepieces so that the eye of an observer can be easily placed in a position coincident with the exit pupil. The eyepieces shall be provided with a diopter scale marked from 0 in plus and minus units. The diopter scale shall have not less than 10 equally spaced graduations in each direction. When focusing on the reticle, the eyepiece shall move without binding or backlash but with a firmness that will retain the focus. The theodolite shall have standard eyepieces of 40X and 24X.

3.5.3 Autocollimation eyepiece. When specified (see 6.2), an autocollimation eyepiece shall be provided for the theodolite. The eyepiece shall be permanently attached to or interchangeable with the telescope eyepiece provided there is no reduction in the optical requirements specified herein or any reduction in magnification.

3.5.4 Eyepiece prism. On theodolites with telescopes that will transit 360 degrees, a prism shall be furnished for the telescope eyepiece. On theodolites with telescopes that will not transit 360 degrees, a prism shall be furnished for the telescope eyepiece and the reading microscope. The eyepiece prism shall give an erect image of the circle and micrometer scales, and the magnification shall be not less than 80 percent of the standard eyepiece. The eyepiece prism shall allow observations of vertical angles of at least 65 degrees above the horizon and reading the circles and micrometer when the telescope is pointed at a target on the ground. The prism shall reverse readily when the telescope is transited.

3.6 Standards. The standards shall be factory adjusted and permanently set. The vertical axis of the theodolite shall be perpendicular to the horizontal axis of the theodolite. With the telescope pointed 30 degrees above the horizontal, then transited through 60 degrees of arc to a point 30 degrees below horizontal, the axis of the telescope shall not be out of perpendicular more than 5 seconds of arc for the theodolites. The error shall not be exceeded with the telescope in either the direct or reverse position.

3.6.1 Horizontal collimation. A line drawn through the axis of the telescope shall be parallel to a line drawn through two points 180 degrees apart on the horizontal circle within an accuracy of 2.5 seconds of arc.

3.6.2 Vertical collimation. With the telescope brought to its level position with the altitude level, a line drawn through the axis of the telescope shall be parallel to a horizontal line drawn through 90 degrees on the left and 90 degrees on the right, or 90 degrees on the left and 270 degrees on the right on the vertical circle within an accuracy of 2.5 seconds of arc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.7 Circles. Horizontal and vertical circles shall be made of optical glass. Graduations and numbers shall be opaque, legible, permanent, and silhouetted against a light background and shall be enclosed in dust-proof and moisture-proof housings.

3.7.1 Horizontal circle. The horizontal circle shall be graduated into 360 degrees at intervals of not more than 10 minutes of arc, and each degree shall be numbered. The numbering shall be from 0 to 359 degrees and shall increase from left to right as the alidade with telescope in the direct position is rotated clockwise. The graduation error of the arithmetic mean of two diametrical lines including the error of observation (maximum deviation or (t) in figure 1) shall be not greater than ±1.0 second. The tolerance of the arithmetic mean of two diametrical lines ((t) in figure 1) shall be ±0.50 second with all eyepieces used for astronomic and triangulation work.

3.7.1.1 Horizontal setting circle. A circle setting device shall be provided for setting the horizontal circle in the various positions of the observation program. The circle setting device shall have free movement without binding throughout complete revolutions of the horizontal circle. Positive means shall be provided to prevent movement of the circle except when desired by the observer.

3.7.2 Circle micrometer. The micrometer shall be graduated at intervals not more than 0.50 second of arc and shall serve for both the horizontal and vertical circles. All movements shall be constant, free, and without backlash throughout the run of the drum.

3.7.3 Microscope. The eyepiece(s) of the micrometer microscope(s) shall be next to the eyepiece of the telescope so that the observer can make all readings without changing his position. The magnification of the horizontal and vertical circles shall be proportional to the diameters of the circles, and the images shall appear as near the same size as practicable. The images of both sides of the horizontal or vertical circles, the index lines, and the micrometer scale shall appear in the same focal plane of the microscope eyepiece. Illumination of the circle images and micrometer scale shall appear uniform and equal in intensity. When focusing on the reading scales, the microscope eyepiece shall move without binding or backlash but with a firmness that will retain the focus.

3.8 Eyepiece filters. Eyepiece filters that can be removed by hand shall be furnished. The telescope eyepiece shall contain a black or nichrome coated neutral filter with a density range of between 5.0 and 6.0. The prismatic eyepiece shall contain a black or neutral filter having a density range of between 5.0 and 6.0, a blue filter having a transmissivity range between 0 and 1, or a red filter having a transmissivity range between 15 and 20; and a green filter having a transmissivity range of between 40 and 45. The coated surface shall be placed adjacent to the telescope eyepiece for protection.

3.9 Levels. Levels shall be provided as specified herein. The material used in marking the level vials and the wear resistance shall conform to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

TT-F-325. Movement of the levels shall be uniform and free from abrupt changes.

3.9.1 Plate level. The plate level shall be graduated to 2 mm spacing and shall have a sensitivity not to exceed 10 seconds, ±2 seconds, for each 2 mm. Graduations shall be clear, uniform, wear-resistant, and symmetrical about the exposed portion and shall extend far enough to include the ends of the bubble at any ambient operating temperature. The level shall be mounted as close to the vertical axis as practicable. The level shall be protected by a transparent cover. A vertical adjusting screw for adjusting the level shall be provided.

3.9.2 Altitude level. The altitude level for indexing the vertical circle shall have a sensitivity not to exceed 12 seconds, ±2 seconds, for each 2 mm. The level shall be read by the coincidence method through a train of prisms and a rotatable viewing glass on the vertical circle standard. The image of the coincident bubble ends shall appear in the viewing glass at any ambient operating temperature. Opposed capstan-headed screws shall be provided for adjustment of the level.

3.9.3 Striding level. A striding level shall be furnished for the theodolites when the rating of the plate level exceeds 7 seconds per 2 mm. The striding level shall have a sensitivity of 0.8 second to 1.5 seconds for each 2 mm graduation and shall be chambered. Provision shall be made for installation of the striding level on the theodolite.

3.9.4 Circular level. The circular level shall have a sensitivity of 12 minutes, ±3 minutes.

3.9.5 Level graduations. The graduations on the levels shall be clear, symmetrical about the exposed portion, wear-resistant, and shall extend far enough to include the ends of the bubble at any ambient temperature from -40 ^{0}F to +125 ^{0}F (-40 ^{0}C to +51.7 ^{0}C).

3.10 Bearings. The rotating alidade shall be supported on a ball bearing. The bearing shall provide precise orientation of the telescope. The bearing shall provide centering with no strain or drag and shall be enclosed in a dust-resistant and moisture-resistant housing.

3.11 Operating screws. Leveling screws and operating screws used in observing shall have heads designed and located to provide manipulation by the operator in total darkness and while wearing arctic gloves. Provision shall be made for wear adjustment of leveling screws and operating screws.

3.12 Optical plummet. The optical plummet shall have a magnification of 2.5 power, ±20 percent. The reticle pattern shall be so placed that it will produce its image concentric with the vertical axis of the instrument.

3.13 Illumination. The theodolite shall be provided with both daylight and electric illumination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.13.1 Daylight illumination. A single, adjustable mirror shall be used to illuminate the vertical and horizontal circles. Mirrors shall be surfaced on the rear face. When silver, magnesium fluoride, or aluminum oxide are used, the mirror shall be second-surfaced on the rear face. Moving or adjusting the reflecting mirror shall not cause a variation in circle reading of more than 0.5 second of arc.

3.13.2 Electric illumination. Lighting attachments shall be provided for the electric illumination of the horizontal and vertical circles, the micrometer scales, the reticles, the altitude level, the striding level, and the autocollimation eyepiece when furnished (see 3.5.3). Lighting attachments shall provide diffused light through frosted glass covers to prevent excess glare to the appropriate eyepiece. Terminal connections shall be to a receptacle on the base of the theodolite.

3.13.3 Reticle illumination. Illumination of the reticle shall be provided which shall retain all the optical requirements specified herein. An adjusting screw or similar device shall be provided for adjusting the brilliance of the reticle illumination.

3.13.4 Wiring. The theodolite shall be wired as required for operation of the lighting equipment. Wiring shall allow sufficient slack between terminals for field repair of broken connections. Slack wire shall not interfere with the operation of any part of the theodolite. An insulated waterproof connector cord with plugs that will fit the theodolite and battery and lighting accessory case receptacles shall be furnished. An insulated, waterproof connector cord not less than 6 feet long (1.8 m) and a plug to fit the battery and lighting accessory case receptacle shall be furnished for the hand lamp specified in 3.13.5. The wire and the connector cord shall be extra flexible and shall not crack or show evidence of damage when wrapped around a 1/4-inch diameter (6.4 mm) mandrel at -40 ^{o}F (-40 ^{o}C).

3.13.5 Hand lamp. A hand lamp with a switch and a connector cord shall be provided.

3.13.6 Electric lamps. Lamps shall be 2.5 volt, 0.30 ampere, miniature screwbase, G-3 1/2 bulb, conforming to W-L-111.

3.13.7 Electric-lamp sockets. Electric-lamp sockets shall be of a size and type to fit 2.5 volt, 0.30 ampere, miniature screwbase, G-3 1/2 bulb, conforming to W-L-111, except for the autocollimating eyepiece illumination which may be standard to the manufacturer.

3.13.8 Lighting accessories. The following accessories shall be furnished in the quantities specified:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Item	Quantity
Connector cord for the base of the theodolite.	1
Connector cord for recording micrometer.	1
Handlamp.	1
Lamps, electric.	Not less than 3

3.14 Tripod. When specified (see 6.2), a rigid, non-extension, leg-type tripod shall be furnished as follows:

a. The legs of the tripod shall be hinged to a metal headplate. The hinges and the headplate shall be of nonmagnetic metal. The hinges shall be adjustable for tension. The legs shall be seated in a metal shoe.
b. The tripod legs shall be made of a seasoned white ash, yellow birch, European birch, hickory, pecan, or hard maple. Slope of the grain of the wood shall not exceed 1 inch (25.4 mm) in 15 inches (381 mm). The wood shall be free of knots.
c. A binding strap of leather or webbing shall be attached to one of the legs near the shoe for holding the legs together when not in use.
d. Unless otherwise specified (see 6.2), a metal protection cap shall be furnished for the head of the tripod.
e. The weight of the tripod shall not exceed 20 pounds (9.07 kilograms (kg)).

3.15 Trivet. Unless otherwise specified (see 6.2), a trivet shall be provided for setting up the theodolite on piers or towers.

3.16 Metal carrying case. A dust-proof and moisture-proof metal carrying case weighing not more than 9 pounds (4.1 kg) shall be furnished. The case shall protect the theodolite against material damage, shifting of optical parts, and damage to controls. A desiccant container shall be provided in the base of the case.

3.17 Transporting case. A reusable transporting case which will house and protect the theodolite in its metal carrying case against rainfall, shock, and vibration during field transportation, shipment, and storage shall be furnished. The case shall be constructed of wood conforming to MIL-STD-731, class 1, groups III and IV. A door capable of repeated use with a hasp and fixed staple shall be furnished. Folding handles for lifting and carrying shall be furnished. Hardware shall be corrosion-resistant or shall be treated to resist corrosion. The case shall weigh not more than 33 pounds (14.97 kg).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.18 Trivet carrying case. Unless otherwise specified (see 6.2) a wooden cushioned carrying case for the trivet shall be provided to protect the trivet against shock and vibration. The case with the trivet shall weigh not more than 20 pounds (9.07 kg).

3.19 Battery and lighting accessories case. A battery and lighting accessories case shall be furnished and shall be wood, leather, or metal. The case shall be attachable to the tripod and shall be constructed so that either six or eight 1.5-volt drycell batteries conforming to MIL-B-18 and MIL-B-18/9, type BA-30, can be used to deliver 3 volts. The case shall be provided with a dimming rheostat and switch. The case shall also accommodate the lighting accessories specified in 3.13.8. The case containing the accessories shall weigh not more than 5 pounds (2.27 kg). Batteries shall not be furnished.

3.20 Winterization kit. When specified (see 6.2), a winterization kit shall be furnished with the theodolite. The winterization kit shall be designed primarily to ensure satisfactory operation of the theodolite and to protect the operator at extremely low temperature (to -25 ^{0}F (-31.7 ^{0}C)). With winterization attachments in place, the theodolite shall fit readily into its carrying case. The kit shall consist of a canvas carrying case containing rubber knob covers and eyepiece shields.

3.20.1 Rubber knob covers. Rubber knob covers shall be provided for all control knobs, leveling screws, and levers. The knob covers shall be of a shape and size to permit manipulation of the controls by an operator wearing heavy mittens. The knob covers shall be rubber conforming to MIL-R-14328 and shall fit snugly.

3.20.2 Eyepiece shields. Eyepiece shields shall be provided for the telescope and the microscope. The shields shall be hard rubber or laminated fabric material. They shall be readily attached or removed without cracking or splitting at any temperature to -50 ^{0}F (-45.6 ^{0}C). The shields shall fit snugly over the eyepieces without disturbing the focus or decreasing the field of view of the telescope or reading microscope. The telescope shall transit with the eyepiece end down when the shields are in place. The shields shall provide protection against frosting of exterior surfaces of the eyepiece lenses without interfering with the accuracy of the instrument. Two sets of eyepiece shields shall be furnished.

3.20.3 Carrying case. The carrying case shall be cotton duck conforming to CCC-C-419, type I, No. 8, color olive drab, and shall be equipped with a belt loop and a slide fastener. The case shall also be provided with a double hook conforming to MIL-H-639, type I, class 1. The case shall be dimensioned and arranged to provide orderly storage for the items contained therein. The following items, in addition to the items listed in 3.20 shall be provided in the carrying case:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Item	Quantity
Lens tissue (NNN-P-40, type I class 1 or 2, 3 by 5 inches).	3 Packages
Clean cloths (washed several times to remove starch and lint) 18 by 18 inches (457.2 by 457.2 mm).	2
Orangewood sticks (commercial grade) (to scrape frost from lenses).	Package of 25
Plastic tape (commercial grade)	1 roll

3.21 Waterproof cover. A lightweight, flexible waterproof cover shall be provided to cover the theodolite when the theodolite is set up for operation.

3.22 Lubrication. Unless otherwise specified (see 6.2), the theodolite shall be lubricated prior to assembly with grease conforming to MIL-G-10924.

3.23 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.23.1), with 1 percent copper 8-quinolinolate (by weight) based on the non-volatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.23.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with the rules associated with volatile organic compounds, as established by local air quality rulings and South Coast Air Quality Management District rulings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

3.24 Identification marking. Each theodolite shall be identified in accordance with MIL-STD-130. When stenciling is used on the canvas cases, the ink shall conform to A-A-208.

3.25 Treatment and painting. The complete instrument, transporting case, battery case, and metal parts of the tripod shall be treated or painted to protect all exposed surfaces against environmental conditions in accordance with the contractor's standard practice. The final coat shall be in accordance with MIL-T-704, type B. The interior of the telescope and optical train shall be matte black. The wooden part of the tripod legs shall not be colored and the grain of the wood shall be visible. The wooden legs shall be treated with a wood preservative surface sealer. The legs shall be finished with varnish in accordance with TT-V-121. Water shall not raise the grain of the wood, affect the finished appearance, or cause the treatment compound to leach from finished compounds when tested.

3.25.1 Mildew-resistant treatment. Unless otherwise specified (see 6.2), cotton webbing, duck, and thread shall be treated for mildew resistance in accordance with MIL-F-46032, type I, grade B, method 1.

3.26 Triangulation and astronomic azimuth.

3.26.1 Weight. The theodolite shall weigh not more than 26 pounds (11.8 kg).

3.26.2 Telescope. The telescope of the theodolite shall not exceed 10.5 inches (266.7 mm) in length and shall transit in the vertical plane to permit observations along a line of sight not less than 65 degrees above the horizon. An elbow eyepiece or prism may be used for such observations.

3.26.3 Vertical circle. The vertical circle shall be graduated according to one of the following patterns, with reference to the face of a clock: 0 degrees at 6 o'clock, 90 degrees at 9 o'clock and 3 o'clock, and 180 degrees at 12 o'clock; or 0 degrees at 12 o'clock, 90 degrees at 9 o'clock, 180 degrees at 6 o'clock, and 270 degrees at 3 o'clock. The instrument shall be graduated at intervals not more than 10 minutes of arc within an accuracy of 2.5 seconds.

3.26.4 Optical system. The optical quality of the telescope shall be such that at night stars of at least the 5th magnitude can be observed. The optical properties shall conform to the following:

Magnification:	
Telescope eyepiece, minimum	40X
Alternate eyepiece, minimum	24X
Effective aperture of objective, minimum	60 mm

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:06Z
Check the source to verify that this is the current version before use.

Field of view, diameter of:	
With telescope eyepiece, minimum	51 minutes
With alternate eyepiece, minimum	1 degree, 40 minutes
Shortest focusing distance:	
With telescope eyepiece, maximum	62 feet (18.9 m)
With alternate eyepiece, maximum	56 feet (17.1 m)

Resolution - The telescope shall clearly resolve high contrast patterns of the National Bureau of Standards Special Publication 374 as specified herein at distances specified.

Minimum magnification	40 diameters Pattern No. 68 Distance - 100 feet (30.5 m)
Minimum magnification	24 diameters Pattern No. 28 Distance - 100 feet (30.5 m)

3.26.5 Telescope eyepiece. The eyepiece of the type I theodolite shall be located adjacent to the telescope so that an observer can make readings without changing his position.

3.26.6 Reticle, telescope. The reticle shall be a series of lines permanently registered on optical glass. The reticle pattern shall consist of a single horizontal line and a vertical line which is perpendicular and centered to the horizontal line. The lower half of the vertical line shall be bifilar and the spacing between the bifilar line shall subtend an apparent angle of 35 seconds, ±5 seconds. All lines shall appear opaque and of uniform width subtending an apparent angle of 2 seconds, ±1 second. The reticle shall be mounted in a metal cell which will not distort when the collimation is adjusted. The collimation adjustment shall be made by two to four easily accessible screws.

3.26.7 Levels. The theodolite shall be equipped with a graduated plate and altitude level, and, when required (see 3.9.3), a striding level shall be furnished. A circular level shall be provided on the alidade and the tribrach or on the tripod and the trivet. Vials shall be ground to uniform curvature and suitably mounted.

3.26.8 Carrying case for levels. A wooden, cushioned carrying case shall be provided to protect the striding level against shock and vibration.

3.27 Workmanship. All movable parts shall move without binding or backlash. Focusing components or assemblies shall retain their focus when

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

finger pressure is released and shall not change focus when transited. Metal surfacings on the optical components shall be free from distortion, and mountings of the prisms and lenses shall not shift. Lenses and prisms shall be free of bubbles, digs, scratches, and other surface markings, and the edges shall be free of chips. Components and assemblies shall show no evidence of forced attachment. Bearings shall show no scratches or irregular surface blemishes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article theodolite shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:06Z
Check the source to verify that this is the currentversion before use.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES DECIMALS ± .010.

4.3.2 Tests. The first article theodolite shall be subjected to the tests marked "X" in column 1 of table 1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined for defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the lot.

4.4.3 Tests.

4.4.3.1 Individual. Each theodolite shall be tested as specified in 4.5.2.5 through 4.5.2.16. Failure of any test shall be cause for rejection of the theodolite.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection of the lot.

4.5 Inspection procedure.

4.5.1 Examination. The theodolite, levels, tripod, carrying case, transporting case, battery case, and accessories shall be examined as specified herein for the following defects:

101. Alidade assembly not as specified (see 3.1).
102. Material not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of materials, material finishes or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the theodolite (see 3.3.4).
107. Telescope will not transit and balance as specified (see 3.5).
108. Optical system does not produce a sharp image, is not sensibly free from internal reflection, spherical and chromatic aberration, coma astigmatism, or distortion (see 3.5.1).
109. Coated optics not as specified (see 3.5.1.1).
110. Eyepieces not as specified (see 3.5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

111. Autocollimation eyepiece not furnished when required (see 3.5.3).
112. Eyepiece prism not furnished when required (see 3.5.4).
113. Circles and micrometer not graduated as specified (see 3.7).
114. Graduations on circles and micrometer not clear and clean-cut or appear fuzzy (see 3.7.2).
115. Filter not as specified (see 3.8).
116. Operating screws not as specified (see 3.11).
117. Illumination system not as specified (see 3.13).
118. Tripod not as specified (see 3.14).
119. Weight of theodolite, tripod, and cases not as specified (see 3.14, 3.16, 3.17, 3.18, 3.26.1).
120. Winterization kit not as specified (see 3.20).
121. Cases not as specified (see 3.20.3).
122. Lubrication not as specified (see 3.22).
123. Fungus and moisture resistance not as specified (see 3.23).
124. Identification marking not as specified (see 3.24).
125. Treatment and painting not as specified (see 3.25).
126. Mildew resistance not as specified (see 3.25.1).
127. Reticle pattern not as specified (3.26.6).
128. Reticle lines not clean-cut, appear fuzzy, or do not appear opaque (see 3.26.6).
129. Striding level not furnished when required (see 3.26.7).
130. Workmanship not as specified (see 3.27).

4.5.2 Tests.

4.5.2.1 Test sequence. The theodolite, levels, tripod, carrying case, transporting case, battery case, and accessories shall be tested as specified herein in accordance with the schedule established in table I. The environmental tests shall be the first tests run on the equipment. The sequence of all other tests is optional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

TABLE I. Test schedule.

First Article	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
X	-	Operational	4.5.2.5	3.4 to 3.26
X	-	Low temperature operation	4.5.2.6	3.4.1
X	-	Low temperature storage	4.5.2.7	3.4.2
X	-	High temperature operation	4.5.2.8	3.4.1
X	-	High temperature storage	4.5.2.9	3.4.2
X	-	Solar radiation	4.5.2.10	3.4.3
X	-	Rainfall	4.5.2.11	3.4.4
X	-	Rainfall test of transporting case	4.5.2.11.1	3.4.4
X	-	Sand and dust	4.5.2.12	3.4.5
X	X	Vibration	4.5.2.13	3.4.6
X	-	Shock	4.5.2.14	3.4.7
X	-	Shock test of transporting case	4.5.2.14.1	3.4.7
X	-	Shock test of tripod	4.5.2.14.2	3.4.7
X	X	Humidity	4.5.2.15	3.4.8
X	-	Standards	4.5.2.16	3.6
X	X	Horizontal collimation	4.5.2.16.1	3.6.1
X	X	Vertical collimation	4.5.2.16.2	3.6.2
X	X	Horizontal circle	4.5.2.17	3.7.1
X	-	Circle micrometer	4.5.2.18	3.7.2
X	-	Level sensitivity	4.5.2.19	3.9 & 3.26.7
X	-	Optical plummet	4.5.2.20	3.12
X	-	Optical adjustment	4.5.2.21	3.13.1
X	-	Vertical circle	4.5.2.22	3.26.3
X	-	Focus	4.5.2.22.1	3.5
		Telescope-optics:		
X	-	Magnification	4.5.2.23.1	3.26.4
X	-	Minimum focus	4.5.2.23.2	3.26.4
X	-	Field of view	4.5.2.23.3	3.26.4
X	-	Effective aperture	4.5.2.23.4	3.26.4
X	X	Resolution	4.5.2.23.5	3.26.4
X	-	Reticle line weight and space between reticle lines	4.5.2.24	3.26.6
X	-	Water resistance (wooden tripod legs)	4.5.2.25	3.25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.5.2.2 Test conditions. Prior to beginning the examination and tests, the theodolite shall be adjusted by the contractor by any established methods to operate as specified herein.

4.5.2.3 Temperature test chamber. The low- and high-temperature tests shall be conducted in a test chamber of a size that will house the theodolite mounted on its tripod, levels, carrying cases, and accessories, and with room around the larger periphery of the tripod legs to make the examination and tests specified herein.

4.5.2.4 Test schedule.

4.5.2.5 Operational test. The theodolite shall be operated and checked as follows to determine its operational capabilities:

a. Actuate all movable parts and check for freedom of movement and non-interference with other parts in various operating positions throughout their maximum ranges.
b. Fit or attach all accessories to the theodolite and check for functionability as specified.
c. Check the levels for free operation, to determine if the bubble remains within the graduations, and if the bubble ends of the coincidence level remain visible and do not run off.
d. Rotate the theodolite about its horizontal and vertical axes, and check the illumination system for uniform and steady operation.
e. Check focus of the microscope for uniform and clear focus around each entire circle.
f. Check parts, components, and accessories for damage.

Evidence that the theodolite, components, or accessories will not function as specified; presence of broken, cracked, or shifted parts on the theodolite or accessories; or opened joints, loose hardware, or splintered, cracked, or swollen wood on the tripod or cases shall constitute failure of this test.

4.5.2.6 Low temperature operation. The assembled theodolite on the tripod with waterproof cover in place, levels, and trivet in their respective cases, carrying case, transporting case, battery case (batteries removed), accessories, and samples of the exterior wire used in wiring the theodolite and the connector cords shall be placed in a temperature control chamber. The temperature in the chamber shall be gradually lowered to -40 ^{0}F (-40 ^{0}C), and maintained at that temperature until the theodolite is in thermal equilibrium. While at this temperature the batteries which are not subject to this test shall be placed in the battery box and the theodolite shall be connected to the battery box. The theodolite shall then be subjected to the operational test specified in 4.5.2.5. The wire samples shall be wrapped around a 0.250 inch (6.4 mm) diameter mandrel. The waterproof cover shall be folded four times. Nonconformance to 3.4.1, or evidence of cracks in wire insulation or the waterproof cover, or broken wire shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.5.2.7 Low temperature storage. After successful completion of the low temperature operating test and maintaining a temperature of -40 °F (-40 °C), disconnect the theodolite from the battery box, remove the batteries from the battery box and remove the batteries and wire samples from the control chamber. The temperature in the control chamber shall then be lowered to -50 °F (-45.6 °C) and maintained at this temperature for 6 hours. The temperature shall then be gradually raised to 79 °F, ±5 °F (26.1°C, ±3°C). The batteries shall then be placed in the battery box and the theodolite connected to the battery box. The theodolite, tripod, levels, cases, and accessories shall be tested as specified in 4.5.2.5. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.8 High temperature operation. The assembled theodolite mounted on the tripod with waterproof cover in place, levels and trivet in their respective cases, carrying case, transporting case, battery case (batteries included), accessories, and samples of wire used in wiring the theodolite and handlamp shall be placed in a temperature control chamber. The temperature in the chamber shall be gradually raised to 125 °F (51.7 °C), and maintained at this temperature until the theodolite is in thermal equilibrium. The theodolite, levels, tripod, cases, and accessories shall then be tested as specified in 4.5.2.5. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.9 High temperature storage. After successful completion of the high operating temperature test, the temperature in the control chamber shall be increased to 155 °F (68.3 °C) and maintained at this temperature for 4 hours. The theodolite, levels, tripod, cases, and accessories shall again be subjected to the test specified in 4.5.2.5. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.10 Solar radiation. The theodolite shall be temperature stabilized at 125 °F (51.7 °C) and a source of directed radiant energy simulating solar radiation of 360 Btu/ft^2/hr shall be applied to the theodolite for a period of 4 hours. At the end of this period, the theodolite shall be tested as specified in 4.5.2.5. In addition, the theodolite shall be observed for peeled, cracked, or blistered paint. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.11 Rainfall test. The theodolite with the eyepiece down shall be placed in a rain testing chamber not less than 36 inches high, 36 inches wide, by 36 inches deep (914 mm by 914 mm by 914 mm). The chamber shall be equipped with water spray nozzles to provide a uniform flow of water over the test area at a rate of 3 inches, ±1 inch, per hour (76.2 mm, ±25.4 mm) measured by a US Weather Bureau type gage. Air velocity shall be provided to simulate wind-driven rain conditions from 0 to 45 degrees. Water temperature shall be 52 °F to 68 °F (11.1 °C to 20 °C). The theodolite shall be subjected to this test for 5 minutes at both 0 degrees and 45 degrees. The presence of water in the telescope, optical system, or on bearings shall constitute failure of this test. The theodolite shall then be placed in the metal carrying case and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

subjected to the test for a period of 30 minutes. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.11.1 Rainfall test of transporting case. The rainfall test of the transporting case may be conducted concurrently with the rainfall test of the theodolite except that the transporting case shall be subjected to the rainfall test specified in 4.5.2.11 for a period of 1 hour. Nonconformance to 3.4.4 shall constitute failure of this test.

4.5.2.12 Sand and dust. The theodolite shall be placed in a sand and dust testing chamber not less than 36 inches high, 36 inches wide, by 36 inches deep (914 mm by 914 mm by 914 mm). The sand and dust (140 mesh silica flour (see 6.4)) shall pass through the chamber at a velocity of 100 to 500 feet per minute (0.51 to 2.54 m per second). The density of sand and dust shall be maintained at 0.1 to 0.5 gram per cubic foot within the test area. The internal temperature shall be maintained at 77 ^{0}F, ±5 ^{0}F (25 ^{0}C, ±3 ^{0}C). The relative humidity shall not exceed 30 percent. The theodolite shall be positioned so as to face into the flow of sand and dust. The theodolite shall be subjected to this test for a period of 15 minutes. The metal carrying case shall be subjected to this test for a period of 1 hour. Nonconformance to 3.4.5 shall constitute failure of this test.

4.5.2.13 Vibration. The theodolite in the metal carrying case shall be fastened on a vibration table in an upright position with web straps. The table shall have a sinusoidal motion that can be controlled within 10 percent of the specified amplitude. The theodolite shall be tested for critical frequencies and resistance between 5 and 52 cycles per second as follows. The theodolite shall be vibrated successively in each of the three mutually perpendicular planes that are parallel to the edges of the metal carrying case over a frequency range of 5 to 52 to 5 cycles per second at a rate that will permit the detection of critical frequencies. The total excursion shall be constant at 0.016 inch (0.406 mm). Critical mechanical frequencies of the theodolite in the metal carrying case shall be determined visually by means of a stroboscopic light, an electronic vibration detection instrument, or by energizing the theodolite and metal carrying case and detecting mechanical vibration through electric indications. The theodolite in the metal carrying case shall not vibrate with a displacement greater than the table input or 0.015 times the table input per foot of unsupported length. The critical points and their respective frequencies shall be recorded. After determining the critical frequency, the theodolite in the metal carrying case shall then be vibrated at each of the critical frequencies for a period of 5 minutes at a total excursion of 0.030-inch (0.8 mm). During each 5 minute period, the frequency shall be adjusted to maintain maximum amplitude of vibration of the equipment. Upon the completion of this test, the theodolite shall be mounted on its tripod and tested for operation as specified in 4.5.2.5. Nonconformance to 3.4.6 shall constitute failure of this test.

4.5.2.14 Shock. The theodolite in the metal carrying case, levels in the carrying case, and trivet in the carrying case shall be flat dropped in an upright position from a height of 6 inches (152.4 mm) onto a level wood floor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

The theodolite shall then be subjected to the operational test specified in 4.5.2.5. The theodolite in its carrying case shall then be placed in the transporting case, and it and the levels in their carrying case and the trivet in its carrying case, after being secured for shipment, shall be dropped from a height of 12 inches (305 mm) so that the bottom edge of the transporting case strikes a level concrete floor when the bottom of the transporting case is inclined 15 to 20 degrees from the horizontal. The theodolite shall again be tested as specified in 4.5.2.5. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.14.1 Shock test of transporting case. A dummy load in the configuration of the metal carrying case and equal in weight to the theodolite and the metal carrying case shall be placed in the transporting case. The transporting case shall be closed and secured as for shipment and subjected to eight drops, one on each bottom edge and one on each corner onto a horizontal concrete floor from a height of 42 inches (1067 mm). The height of drop shall be measured from the concrete floor to the nearest edge or corner of the transporting case when suspended prior to each drop. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.14.2 Shock test of tripod. With the protection cap in place, the tripod shall be held in a vertical position 12 inches (305 mm) above a wood floor, dropped, and allowed to fall from the vertical to a horizontal position on the wood floor. Nonconformance to 3.4.7 shall constitute failure of this test.

4.5.2.15 Humidity. Place the theodolite with power off, mounted in the operating position with all accessories, in a humidity chamber conforming to MIL-STD-810, method 507.2, procedure II. The theodolite shall be inspected and tested as specified in 4.5.2.5. Evidence of damage shall constitute failure of this test.

4.5.2.16 Standards. The standards test shall be as follows:

a. With the telescope in the direct position, transit the telescope to 30 degrees above the horizontal and establish a target, A.
b. Transit the telescope to 30 degrees below the horizontal and establish a target, B.
c. Read the horizontal circle.
d. Transit the telescope to the reverse position and sight the telescope on target, A.
e. Transit the telescope to 30 degrees below the horizontal and establish a target, C.
f. Read the horizontal circle.
g. Add 180 degrees to the smaller reading.
h. The difference between the larger and the smaller is twice the error. Divide by 2.

Nonconformance to 3.6 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.5.2.16.1 Horizontal collimation. The horizontal collimation error of the line of sight shall be determined by comparing direct and reverse readings on any one collimator. The error is determined as follows:

a. With the telescope in the direct position, sight on the reticle in the collimator and read the horizontal circle (e.g., 191°01'58").
b. Transit the telescope to the reverse position and sight on the reticle in the collimator and read the horizontal circle (e.g., 11°02'03"). Add 180 degrees to this reading (total 191°02'03").
c. Subtract the smaller reading from the larger reading (e.g., 191°02'03" minus 191°01'58" equals 5 seconds). The 5 seconds is twice the collimation error.

Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.16.2 Vertical collimation. The vertical collimation error of the line of sight shall be determined by direct and reverse readings of the vertical circle with the horizontal crossline. The error is determined as follows:

a. Bring the vertical circle into the field of the microscope with the changeover knob.
b. With the telescope in the direct position, level the telescope with the altitude level.
c. Sight the telescope on a suitable target and read the vertical circle. Make a note of the difference in this reading and 90 degrees. The difference shall not exceed 2.5 seconds of arc.
d. Transit the telescope to its reverse position and sight the telescope on a target (see c. above). Read the vertical circle and make a note of the difference in this reading and 270 degrees or 90 degrees.
e. The difference obtained in c. added to the difference obtained in d. is twice the error.

Nonconformance to 3.6.2 shall constitute failure of this test.

4.5.2.17 Theodolite system (horizontal). The test shall be reported on a test report form similar to that shown in figure 1. A duplicate of the complete test report form shall be placed within the carrying case of the instrument to which the report applies. The test shall be conducted in accordance with the following instructions which shall be printed as part of the test report form:

This test is based on a series of observations on two collimators approximately 36 degrees apart and on a plane having an elevation approximately the same as that of the horizontal axis of the instrument being tested. In making the following observations, each circle reading shall be estimated to the nearest tenth of a graduation of the breakdown scale. This test measures the following parameters of the theodolite:

a. Ability of the instrument to repeat measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

b. Average circle graduation error.
c. Collimation error.

Two sets of measurements will be made. Each set will consist of 20 angles (40 measurements) by using all sections of the horizontal circle as follows:

a. With the telescope in the direct position, the instrument is pointed on the left collimator with a circle reading of approximately 0 degrees and 36 minutes. The circle is read and the seconds are recorded on line 1, column I. The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column I. The telescope is then reversed and pointed on the left collimator, the circle read, and the seconds recorded in line 1, column II (this will be 180 degrees rather than 0 degrees as with the telescope direct). The angle is then turned to the right collimator, the circle read, and the seconds recorded on line 2, column II. The mean of the two readings (columns I and II) is entered in column III. Repeat this technique by changing the position of the horizontal circle to read between 36 and 72 degrees. Advance the circle in this fashion until all parts of the circle are covered during the test.

b. The second set of measurements shall be made identically and shall be recorded in columns IV and V. The mean of these readings shall be in column VI. Taking the difference between the two mean readings from column III, angle 1 shall be determined and shall be written in column VII. Likewise the difference between two mean readings from column VI shall be made and angle 2 shall be determined and written in column VIII. The difference of these two angles represents the pointing and reading error of the test (Δ). These shall be written in column IX. Column X is Δ^2. From column X, $M_2^2 = \frac{\Sigma\Delta^2}{n}$ shall be calculated, where n represents the total number of readings. In the case of 36 degrees, n = 40; therefore $M_2^2 = \frac{\Sigma\Delta^2}{40} \leq 4 \text{ sec}^2$.

Numbers for larger than 4 sec^2 indicate excessive pointing and reading error. The test setup should be checked and, after correction, two new complete sets of readings shall be made. Excessive values of M_2^2 the second time indicate that the instrument is capable of accurately repeating readings and shall constitute failure of this test.

If M_2^2 is smaller than 4 sec, proceed with calculations as follows:

Means of the two angles are determined from columns VII and VIII and entered in column XI, mean. These are summed and mean is again determined. Residuals are found and entered in column XII (V). Column XII shall be scanned. Maximum plus and minus deviations shall be determined and underlined. Any value in excess of 1.0 sec shall constitute failure of this test. If plus or minus

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:06Z
Check the source to verify that this is the current version before use.

maximum value is not excessive, compute mean (V). Also compute V^2 and enter in column XIII. Zero is entered opposite the first residual from column XII and the algebraic sum of the residuals is entered in column XIV. These values are added algebraically and a mean is determined. This mean is subtracted algebraically from the values of column XIV to give column XV(t). The summation of the column should equal zero. Column XV represents the systematic error of the instrument. Column XVI is t2. Column XVII is the difference between column I and column II ($D_1 - R_1$). The sum of this column divided by twice the amount of readings equals the collimation error. In the case of 36 degree collimators, $C = \frac{\Sigma(D_1 - R_1)}{40} \leq$ 12 sec. Any value exceeding 12 sec constitutes failure of this test.

The circle graduation error (t) is computed as follows:

$$M_1^2 = \frac{\Sigma V^2}{n_c - 1} = \frac{\Sigma V^2}{9}$$

$$M_2^2 = \frac{\Sigma \Delta^2}{n_R} = \frac{\Sigma \Delta^2}{40}$$

$$2t^2 = M_1^2 - M_2^2$$

if $M_1^2 - M_2^2 \geq 0$

then $t = \sqrt{\frac{M_1^2 - M_2^2}{2}} \leq 0.08$ min (0.024 *mils*)

Exceeding this value constitutes failure of this test.

if $M_1^2 - M_2^2 < 0$

then compute $(t)m^2 = \frac{\Sigma (t)}{n_c} = \frac{\Sigma (t)}{10}$

$$(t)m = \sqrt{\frac{\Sigma (t)}{10}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Note: (t)m is not the true circle graduation error; it is the average systematic that includes the circle graduation error.

Since $M_2^2 \leq 4$ sec^2 was earlier established, we know that the t ≤ 0.5 sec. This is only true if $M_1^2 - M_2^2 < 0$.

The number of angles is represented by $n_<$ and n_R is the number of readings.

4.5.2.18 Circle micrometer. The run of the micrometer shall be tested by making coincidence between graduations on opposite sides of the circle with the micrometer reading approximately 0. The exact reading is recorded. The micrometer knob shall then be turned until coincidence is again made near the 60 second end of the micrometer, and the exact micrometer reading shall be recorded. Three sets of readings in each quadrant of the circles shall be made. The mean reading for each end of the micrometer scale shall be calculated. Nonconformance to 3.7.2 shall constitute failure of this test.

4.5.2.19 Level sensitivity. The sensitivity of level vials shall be determined by mounting the level vials on a level tier or by any other method yielding values accurate within 5 percent. The plane of the graduated vials shall be changed the entire amount, the bubble shall be driven over the whole step-by-step first in one direction and then the other, and the plane of the ungraduated vials shall be changed a sufficient amount to cause a change of not less than 4 mm in the position of the level bubble as indicated on a scale placed adjacent to the vials. Nonconformance to 3.9 shall constitute failure of this test.

4.5.2.20 Optical plummet. The optical plummet shall be operated to view the image of a hub tack at tripod height. Nonconformance to 3.12 shall constitute failure of this test.

4.5.2.21 Mirror adjustment. The mirror shall be adjusted both vertically and horizontally while the micrometer is observed. Nonconformance to 3.13.1 shall constitute failure of this test.

4.5.2.22 Vertical circle. The vertical circle shall be tested during the production of the theodolite. Nonconformance to 3.26.3 shall constitute failure of this test.

4.5.2.22.1 Focus. Focus collimation error shall be determined by testing as for the horizontal collimation test except that targets at 2 feet (.61 m) and at infinity in the same line of sight shall be used. The results are reduced as specified in the horizontal collimation test. The difference between the error at 62 feet (18.9 m) and the error at infinity is the focus collimation error. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.23 Telescope-optics.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:06Z
Check the source to verify that this is the current version before use.

4.5.2.23.1 Magnification. The magnifying power, in diameters, of the telescope shall be measured by focusing the telescope for parallel light with the aid of a low-power auxiliary telescope placed behind the eyepiece of the telescope under test. Without change of focus, a transparent graduated scale shall be placed in front of the objective lens and as near thereto as practicable. The low-power telescope shall be replaced with a measuring microscope and focused on the image of the scale. The magnification of the telescope is then determined in diameters as the ratio given length of the scale to the length of its image in the exit pupil. Any other method yielding values accurate to within 2 percent may be used in lieu of the specified method. Evidence of a magnification for the telescope less than that specified in 3.26.4 shall constitute failure of this test.

4.5.2.23.2 Minimum focus. The telescope shall be set at minimum focus. A graduated scale or object marked with sharply-defined lines shall be moved along the line of sight from the telescope until its markings are clearly visible by the lens of the telescope. The shortest distance from the telescope to a point on the line of sight at which the markings become clearly visible is the minimum focus value. This distance shall be measured. Nonconformance to 3.26.4 shall constitute failure of this test.

4.5.2.23.3 Field of view. The horizontal width of the field of view shall be measured by the use of two targets. The two targets shall be placed 500 m from the theodolite. One target shall be placed on a horizontal crossline at the extreme left of the field of view. The second target shall be placed on the horizontal crossline at the extreme right of the field of view. The angle between the two targets shall then be measured. The vertical width of the field of view shall be measured in the same manner except that the two targets shall be set on a vertical crossline. Nonconformance to 3.26.4 shall constitute failure of this test.

4.5.2.23.4 Effective aperture. A ruled glass scale shall be placed in contact with the front of the objective glass. By sighting through the eyepiece, the visible portion of the scale shall constitute the effective aperture. Nonconformance to 3.26.4 shall constitute failure of this test.

4.5.2.23.5 Resolution. Observations for resolution shall be made by employing charts contained in National Bureau of Standards Special Publication 374. The telescope shall be set up 100 feet (30.48 m) from pattern 68 shown on the NBS chart. Nonconformance to 3.26.4 shall constitute failure of this test.

4.5.2.24 Reticle line weight and space between reticle lines. The reticle line weight and the space between reticle lines shall be determined by sighting on a collimator with a target in angular units or any other target that will yield comparable values. Three readings shall be taken for each test. The mean of the three readings shall be calculated. Nonconformance to 3.26.6 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

4.5.2.25 Water resistance (wooden tripod legs). This test shall be conducted in the manufacturer's plant during the production of the tripods. Not less than three samples treated and varnished in accordance with 3.25 and allowed to dry for a period of 168 hours shall be placed upright to half their length in water at room temperature and allowed to stand for 24 hours. Surface moisture shall then be removed with paper toweling. The samples shall then be examined for raising of the grain of the wood, loss of transparency, and evidence of stain on the paper toweling from the sealer or varnish. Nonconformance to 3.25 shall constitute failure of this test.

4.6 Inspection of packaging. Level A, level B or level C preservation, packing, and marking shall be examined and tested in accordance with the quality assurance provisions of MIL-S-3534.

5. PACKAGING

5.1 Preservation packing and marking. Preservation, packing, and marking shall be in accordance with levels A, B or C requirements of MIL-S-3534 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The theodolite with accessories is intended for use in precise geodetic surveying and for triangulation and astronomic work where first-order results are required.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of first article model and number of theodolites required (see 3.2).
d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
e. When an autocollimation eyepiece is required (see 3.5.3).
f. When tripod shall not be furnished (see 3.14).
g. When metal cap shall not be furnished (see 3.14).
h. When trivet shall be furnished (see 3.15).
i. When carrying case for the trivet is not required (see 3.18).
j. When a winterization kit is required (see 3.20).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

k. When the theodolite shall not be lubricated prior to assembly (see 3.22).
l. When mildew resistant treatment shall be other than as specified (see 3.25.1).
m. Level of preservation, packing and marking required (see 5.1).
n. When composition II is required (see 3.23).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Silica flour. The sand and dust to be used in the sand and dust test is commercially known as "140-mesh silica flour". Sand and dust, 140 silica flour, produced by the Fenton Foundry Supply Company, Dayton, Ohio, and Ottawa Silica Company, Ottawa, Illinois, or equal have been found satisfactory.

6.5 First order. The probable error allowed for first order astronomic observations is: latitude, ±0.2 second, longitude, ±0.2 second; azimuth for triangulation, ±0.3 second, and azimuth for traverse, ±0.5 second. More complete information regarding first order will be found in Department of the Army Technical Manual, TM 5-441, Topographic Surveying.

6.6 Horrebow-Talcott level. As methods change from the Horrebow-Talcott method for determining latitude to methods requiring the use of the vertical circle, the accuracy of the vertical circle becomes of more importance.

6.7 Technical Manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be listed on a separate Contract Data Requirement List (DD form 1423), which is included as an exhibit to the contract. The Technical Manual must be required under separate contract line item in the contract.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of theodolites.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

6.9 Metric values. Metric values shown in parenthesis are provided for information only.

6.10 Subject term (keyword) listing.

a. Alidade
b. Geodetic surveying
c. Microscopic optical system
d. Triangulation

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - GS

Project 6675-A937

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

Column		I	II		III	IV	V		VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII
Pointing		Mils Min D_1 Sec	Mils Min R_1 Sec		$\frac{D_1+R_1}{2}$	Mils Min D_2 Sec	Mils Min R_2 Sec		$\frac{D_2+R_2}{2}$	<1 (B-A)	<2 (D-C)	(<1 - <2)	s	$\frac{<1 + <2}{2}$	V (XI - M_1)	V^2	R ($V_1+V_2+V_3$)	(t) (XIV - M)	$(t)^2$	D_1-R_1
Deg	mil																			
0	0			A				C												
36	640			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
36	640			A				C												
72	1280			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
72	1280			A				C												
108	1920			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
108	1920			A				C												
144	2560			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
144	2560			A				C												
180	3200			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
180	3200			A				C												
216	3840			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
216	3840			A				C												
252	4480			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
252	4480			A				C												
288	5120			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
288	5120			A				C												
324	5720			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
324	5720			A				C												
0	0			B				D		XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	XXX	
														I R =	I V_3 =		I R =		I C =	
												$M_2^2 = \frac{I^2}{48}$ =		$M_1 = \frac{I R}{10}$ =	$M_3^2 = \frac{I V^2}{9}$ =		$M_2 = \frac{I R}{10}$ =		$C = \frac{I C}{40}$ =	

$2t^2 = M_1^2 - M_2^2$ if $M_1^2 - M_2^2 \geq 0$ then $t = \pm\sqrt{\frac{M_1^2 - M_2^2}{2}}$

Mfgr _____ Date _____

Ser. No. _____

Order _____

Tested by _____

Checked by _____

FIGURE 1. Data Sheet for Theodolite System Test (horizontal). X-3034

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-52035C (ME)	91/04/25

3. DOCUMENT TITLE
Theodolites, Surveying, Direction, First-Order, with Accessories, Tripod, and Case

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 664-5717 (2) AUTOVON: 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466

Source: https://assist.dla.mil -- Downloaded: 2026-01-26T04:06Z
Check the source to verify that this is the current version before use.